

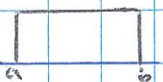
CONTINUOUS RANDOM VARIABLES (CRVs)

A continuous random Variable (X) has a **Probability Density Function** $f(x)$ p.d.f (Probs. Density Function)

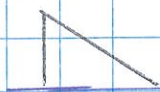
e.g.



normal



rectangular
or
uniform



triangular

etc

All probability density functions have these properties:

1. Total area = 1

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

2. Always ≥ 0

$f(x) \geq 0$ for all values of x

3. Probabilities can be found by calculating areas using integration (or geometry)

$$P(a < x < b) = \int_a^b f(x) dx$$

Example.

X has p.d.f $f(x)$ where

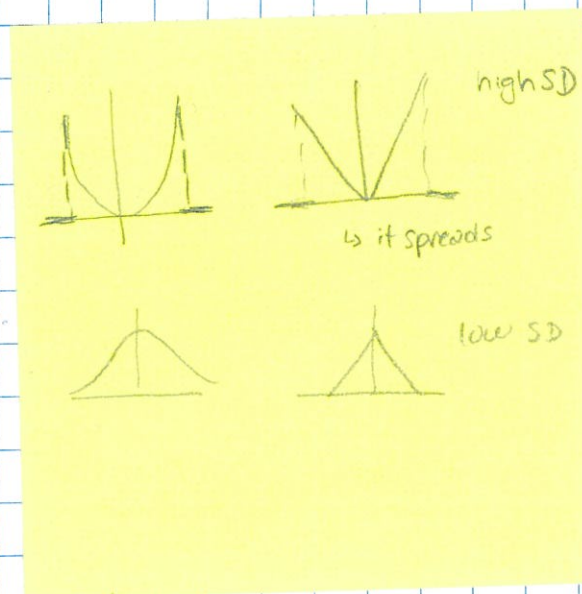
$$f(x) = \begin{cases} K(1+x^2) & \text{for } -1 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

where K is constant.

1. Sketch $f(x)$

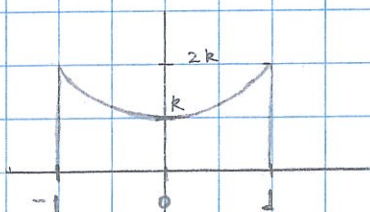
2. Find the value of K

3. Find $P(0.3 \leq X \leq 0.6)$



Ans:

1. if $x = -1$ $f(x) = 2k$
 $x = 0$ $f(x) = k$
 $x = 1$ $f(x) = 2k$



2. $\int_{-1}^1 f(x) dx = 1$ so, $\int_{-1}^1 k(1+x^2) dx = 1$

$$k \int_{-1}^1 1+x^2 dx = 1$$

$$k \left[x + \frac{x^3}{3} \right]_{-1}^1 = 1$$

$$\left[1 + \frac{1}{3} \right] - \left[-1 - \frac{1}{3} \right] = \frac{1}{k}$$

$$\frac{4}{3} + \frac{4}{3} = \frac{1}{k}$$

$$k = \frac{3}{8}$$

3. $P(0.3 \leq x \leq 0.6)$

$$\frac{3}{8} \int_{0.3}^{0.6} 1+x^2 dx$$

$$= \frac{3}{8} \left[x + \frac{x^3}{3} \right]_{0.3}^{0.6} = \frac{3}{8} \left[\left(0.6 + \frac{(0.6)^3}{3} \right) - \left(0.3 + \frac{(0.3)^3}{3} \right) \right]$$

$$= \frac{3}{8} \left[\frac{84}{125} - \frac{309}{1000} \right] = \frac{3}{8} \times \frac{363}{1000}$$

$$= \frac{1089}{8000} = 0.136125$$

$$= 0.136$$

Continuous Random Variables

Total area, median and quartiles (not given on formula sheets):

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\int_{-\infty}^M f(x) dx = 0.5$$

$$\int_M^{\infty} f(x) dx = 0.5$$

$$\int_{-\infty}^L f(x) dx = 0.25$$

$$\int_U^{\infty} f(x) dx = 0.75$$



where M = median, L = lower quartile, U = upper quartile

Mean and variance (given on formula sheets):

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

multiply then
integrate

$$\text{Var}(X) = \int_{-\infty}^{\infty} x^2 f(x) dx - \{E(X)\}^2$$

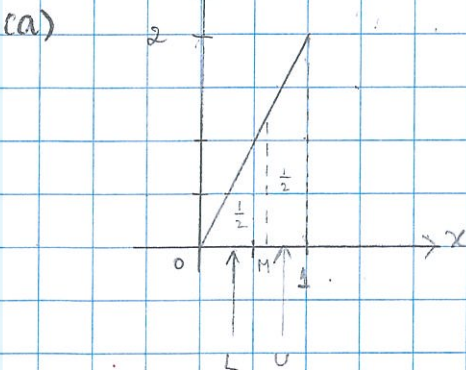
NOTE: The interquartile Range

$$\text{IQR} = \text{UQ} - \text{LQ}$$

E.g. $f(x) = \begin{cases} 2x & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$

(a) sketch $f(x)$

(b) find median, LQ, UQ, IQR



$$(b) \int_0^M 2x dx = 0.5$$

$$[x^2]_0^M = 0.5$$

$$[M^2] - [0^2] = 0.5$$

$$M^2 = \frac{1}{2}$$

$$M = \frac{1}{\sqrt{2}}$$

LQ

$$\int_0^L 2x \, dx = 0.25$$

$$\int_0^L [x^2]_0^L = 0.25$$

$$L^2 - 0^2 = \frac{1}{4}$$

$$L = \frac{1}{2}$$

UQ

$$\int_0^U 2x \, dx = 0.75$$

$$[x^2]_0^U = 0.75$$

$$U^2 - 0^2 = 0.75$$

$$U = \sqrt{0.75}$$

$$U = \frac{\sqrt{3}}{2}$$

$$IQR = UQ - LQ = \frac{\sqrt{3}}{2} - \frac{1}{2} = \frac{\sqrt{3}-1}{2}$$

(c) Find mean and variance $E(X)$, $\text{var}(X)$ 

long term average

$$0.35 + 0.4 + \frac{0.412 + 0.6 + 0.912}{5}$$

$$E(X) = \int_0^1 x f(x) \, dx = \int_0^1 x (2x) \, dx = \int_0^1 2x^2 \, dx$$

$$= \left[\frac{2}{3} x^3 \right]_0^1 = \left[\frac{2}{3} \right] - [0] = \frac{2}{3}$$

$$\text{Var}(X) = \int_0^1 x^2 f(x) \, dx - \{E(X)\}^2$$

$$= \int_0^1 x^2 (2x) \, dx - \left(\frac{2}{3} \right)^2$$

$$= \int_0^1 2x^3 \, dx - \frac{4}{9}$$

$$= \left[\frac{2x^4}{4} \right]_0^1 - \frac{4}{9} = \left[\frac{x^4}{2} \right]_0^1 - \frac{4}{9}$$

$$= \left[\frac{1}{2} - \frac{0}{2} \right] - \frac{4}{9} = \frac{1}{2} - \frac{4}{9} = \frac{1}{18}$$